

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.6	Gender	Male
Height (cm)	175.0	Weight (kg)	85.0
BMI	27.8	Product Info 2	A1
Occupation	Pension	Annual Income	\$400,000.00
Insurance Type	Life	Coverage Amount	\$500,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	Low Risk
Recommended Annual Premium	\$10,000.00

3. AI Actuarial Analysis

Actuarial Report: Insurance Application Evaluation

Application ID: N/A (Current Applicant Details)

Date: 2023-10-27

Product Type: Life Insurance

Coverage Amount: \$500,000.00

1. Risk Assessment

The applicant's profile presents several critical inconsistencies and high-risk factors that warrant immediate attention:

* **Age (0.6 years):** This indicates an infant. This is the most significant outlier as it conflicts severely with almost all other provided attributes.

* **BMI (27.8 kg/m²), Height (175.0 cm), Weight (85.0 kg):** These physical dimensions and BMI are characteristic of an adult, specifically indicating an overweight or borderline obese individual. This is physically impossible for an infant aged 0.6 years.

* **Smoker (1):** An infant cannot be a smoker. This is a direct contradiction with the age.

* **Occupation ('Pension') & Income (\$400,000.00):** An infant cannot have an occupation or an income of this magnitude. 'Pension' typically implies retirement, which would fit an older individual.

* **Lifestyle (Exercise: 0, Alcohol: 3):** These lifestyle indicators, while subjective, are also inconsistent with an infant's profile.

* **Previous Claims (0) & Family History (0):** These factors are favorable, indicating no adverse claims history or family predispositions, but are overshadowed by the other inconsistencies.

* **Gender (Male):** Neutral.

* **Product Info 2 (A1):** Neutral without further context.

Summary of Key Risk Factors: The entire profile is compromised by a fundamental inconsistency between the applicant's stated age and nearly all other demographic, health, and financial indicators. If the age were accurate, the other data points are impossible. If the other data points were accurate (suggesting an adult), then the age is erroneous. This data integrity issue prevents a reliable risk assessment.

2. Comparative Analysis

The Machine Learning Classifier output and RAG context indicate a predicted Risk Class 1 with a calculated annual premium of \$10,000.00. The top three similar historical cases (APP-0642B5, POL-59031, POL-46791) show 100% similarity, the same Risk Class 1, and the same premium, all of which were approved.

Comparison:

The current application *numerically matches* the features of these historical approved cases, implying that the raw numerical values (e.g., BMI, income, smoker status, previous claims) align with a profile previously deemed low risk. The high similarity score and identical output suggest the ML model successfully found a direct match in its training data or RAG context.

****Differences:****

The critical difference lies in the ****semantic interpretation and internal consistency**** of the data. While the numbers might match, the ***combination*** of `age` 0.6 with adult physical attributes, occupation, income, and smoker status, is an internal contradiction not present in a truly valid "Risk Class 1" profile. The historical cases, though numerically identical in features, are presumed to be for ***valid*** applicants where such contradictions do not exist. The classifier likely processed the numerical inputs without fully comprehending the biological and logical impossibilities presented by the combination of features.

3. Anomaly & Fraud Detection

This application exhibits severe anomalies, strongly suggesting a data entry error or potential misrepresentation:

*** **Gross Age Inconsistency:**** An age of 0.6 years (infant) is fundamentally incompatible with a height of 175cm, weight of 85kg, and BMI of 27.8.

*** **Biological Impossibilities:**** An infant cannot be a smoker (smoker: 1) nor have an occupation ('Pension') or an income of \$400,000.00.

*** **Lifestyle Indicators:**** The exercise (0) and alcohol (3) levels are adult lifestyle factors, not applicable to an infant.

*** **High Coverage for Age:**** A \$500,000 life insurance policy on an infant, while not an anomaly in itself, raises questions regarding insurable interest when combined with the other highly irregular data.

The combination of these factors points to a severe data integrity issue. It is highly improbable that these details accurately describe a single individual. This profile's inconsistencies warrant immediate investigation before any underwriting decision.

4. Premium Recommendation

The calculated annual premium of \$10,000.00, derived from a predicted Risk Class 1, cannot be justified given the current application details.

*** The Machine Learning Classifier's** confidence level of 55.5% is relatively low for a predicted "best" risk class (Class 1), which typically sees much higher confidence. This lower confidence, despite the low-risk classification, might subtly reflect the model's uncertainty or inability to reconcile underlying feature relationships.

*** Due to the fundamental data inconsistencies** detailed in the "Anomaly & Fraud Detection" section, any premium derived from these inputs is unreliable and invalid. If the age is genuinely 0.6, the other data is false. If the other data is true (adult applicant), the age is false, and the actual risk profile (e.g., adult smoker with BMI 27.8) would lead to a different premium calculation.

Therefore, the premium of \$10,000.00 is ****not justifiable**** based on the current, flawed data.

5. Underwriting Decision Recommendation

****Recommendation: Refer for Manual Review****

Given the severe and pervasive data inconsistencies, this application cannot be approved or rejected based on

the provided information. It requires immediate manual intervention and verification.

****Actions required during manual review:****

1. ****Verify Applicant Identity and Data:**** Confirm the applicant's true age, height, weight, BMI, smoker status, occupation, income, and lifestyle factors.
2. ****Identify Data Entry Error:**** Determine if the age was a data entry error (e.g., 0.6 entered instead of 60, or missing a decimal, etc.), or if the other demographic data belongs to a different individual.
3. ****Assess Misrepresentation:**** Investigate if there is any deliberate misrepresentation of facts.
4. ****Recalculate Risk:**** Once accurate data is obtained, re-evaluate the risk profile and recalculate the premium accordingly.

Until these discrepancies are resolved, the application is un-underwritable.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.